

Checklist

Disaster Recovery

1. Define Recovery Objectives

- ☐ Identify critical business services, applications, systems, and data.
- ☐ Define the **Recovery Point Objective (RPO)** for each critical system.
- ☐ Define the **Recovery Time Objective (RTO)** for each critical system.
- ☐ Confirm that RPO and RTO targets are driven by business requirements.
- ☐ Review the cost and technical complexity of meeting each target.
- ☐ Obtain agreement from business owners on acceptable recovery targets.
- ☐ Prioritize systems according to business impact and recovery urgency.

2. Select Recovery Strategies

- ☐ Determine which recovery approach is appropriate for each system.
- ☐ Consider **traditional physical backups** where appropriate.
- ☐ Store backup media at a separate, secure location.
- ☐ Confirm replacement hardware or infrastructure will be available after a disaster.
- ☐ Factor backup transport and restoration time into the RTO.
- ☐ Consider a **warm standby** environment for faster recovery.
- ☐ Keep standby systems reasonably synchronized with production.
- ☐ Keep standby systems geographically separated from the primary environment.
- ☐ Document the process for redirecting DNS, routing, or network traffic during failover.
- ☐ Consider **high availability** for systems requiring very short RPOs and RTOs.
- ☐ Ensure high-availability environments have enough spare capacity after a failure.
- ☐ Consider an **alternate system** when restoring the original system is unnecessary.
- ☐ Consider **system function reassignment** by repurposing noncritical systems during a disaster.
- ☐ Verify reassigned systems can safely support production workloads.

3. Plan Cloud-Native Disaster Recovery

- ☐ Replicate critical applications and data to another cloud region or location.
- ☐ Determine whether replication should be continuous or scheduled.
- ☐ Use **Infrastructure as Code (IaC)** to recreate recovery environments quickly.
- ☐ Verify IaC configurations accurately reproduce the production environment.
- ☐ Configure cloud resources to scale when disaster recovery is activated.
- ☐ Consider multi-region deployment for critical services.
- ☐ Configure traffic rerouting to healthy regions.
- ☐ Automate failover and failback where possible.
- ☐ Use orchestration tools to reduce manual recovery steps.
- ☐ Monitor cloud replication and recovery readiness.
- ☐ Regularly test cloud recovery procedures.
- ☐ Review cloud DR costs during normal and disaster operations.

4. Identify Dependencies

- ☐ Map dependencies for every critical application and service.
- ☐ Identify required network connectivity.
- ☐ Identify routing and DNS dependencies.
- ☐ Identify authentication and directory-service dependencies.
- ☐ Identify databases, storage, APIs, and external services required by each system.
- ☐ Confirm dependent services have compatible RTO and RPO targets.
- ☐ Adjust unrealistic recovery targets when dependencies cannot meet them.
- ☐ Include dependencies in tabletop exercises and DR testing.

5. Prepare Disaster Scenarios

- ☐ Develop a ransomware or cyberattack scenario.
- ☐ Develop a mission-critical hardware failure scenario.
- ☐ Develop a complete datacenter or site-loss scenario.
- ☐ Develop a prolonged power-outage scenario.
- ☐ Develop a fire, flood, earthquake, or other physical-disaster scenario.
- ☐ Develop a pandemic or loss-of-site-access scenario.
- ☐ Include representatives from relevant IT and business teams when reviewing scenarios.

- ☐ Identify how each scenario affects systems, people, facilities, and communications.
- ☐ Determine whether the existing DR plan adequately addresses each scenario.

6. Define Failover Procedures

- ☐ Establish clear criteria for declaring a disaster.
- ☐ Define who is authorized to activate the DR plan.
- ☐ Establish an escalation process for uncertain situations.
- ☐ Document the exact steps required to fail over critical services.
- ☐ Identify who performs each failover action.
- ☐ Document DNS, routing, application, and infrastructure changes required during failover.
- ☐ Confirm failover can meet the defined RTO.
- ☐ Maintain an emergency communication plan for staff and stakeholders.

7. Define Failback Procedures

- ☐ Document how services will return to the primary environment.
- ☐ Confirm the original environment is stable before failback.
- ☐ Synchronize data before returning services to the primary environment.
- ☐ Plan failback during an appropriate maintenance window where possible.
- ☐ Identify who is authorized to approve failback.
- ☐ Communicate planned service impacts before switching back.
- ☐ Verify applications and services after failback.
- ☐ Document and resolve any issues discovered during the process.

8. Test the Disaster Recovery Plan

- ☐ Schedule regular DR tests.
- ☐ Perform tabletop exercises.
- ☐ Perform technical recovery exercises.
- ☐ Test scenarios without relying on systems that would be unavailable during the simulated disaster.
- ☐ Verify systems can be recovered within the RTO.
- ☐ Verify data can be recovered within the RPO.
- ☐ Test backup restoration rather than assuming backups are usable.
- ☐ Test network, DNS, authentication, and application dependencies.

- ☐ Observe and document each test.
- ☐ Record what worked well.
- ☐ Record failures, delays, and unexpected dependencies.
- ☐ Conduct a post-test debrief.
- ☐ Assign corrective actions to specific owners.
- ☐ Set deadlines for corrective actions.
- ☐ Update the DR plan based on lessons learned.
- ☐ Retest major changes.

9. Protect Data at Rest

- ☐ Apply appropriate security controls to replicated and backup data.
- ☐ Encrypt sensitive backup data where appropriate.
- ☐ Restrict access to backup and recovery systems.
- ☐ Require strong authentication for administrative access.
- ☐ Apply controls comparable to those used in the production environment.
- ☐ Confirm backup copies do not create a weaker path to sensitive information.

10. Protect Data in Transit

- ☐ Encrypt data transmitted to backup or secondary systems.
- ☐ Authenticate systems participating in replication.
- ☐ Secure network connections used for backup and recovery.
- ☐ Review replication traffic for exposure to untrusted networks.
- ☐ Monitor recovery-related network activity for suspicious behavior.

11. Maintain Patching and Configuration

- ☐ Keep recovery systems patched at an appropriate level.
- ☐ Keep production and recovery configurations aligned.
- ☐ Apply configuration changes to both primary and recovery environments.
- ☐ Track intentional differences between environments.
- ☐ Verify applications behave correctly on recovery systems.
- ☐ Prevent outdated recovery systems from becoming a security weakness.

12. Control User Access

- ☐ Define who can access recovery systems during a disaster.
- ☐ Maintain least-privilege access wherever possible.
- ☐ Protect personally identifiable, financial, and other sensitive information.
- ☐ Do not remove security controls simply to accelerate recovery.
- ☐ Document emergency-access procedures.
- ☐ Review and revoke temporary access after the incident.

13. Maintain Physical Security

- ☐ Confirm secondary sites have appropriate physical access controls.
- ☐ Compare physical security at primary and recovery locations.
- ☐ Protect servers, storage, backup media, and networking equipment.
- ☐ Restrict access to authorized personnel.
- ☐ Address security gaps if a secondary location uses office space rather than a dedicated datacenter.

14. Communication and Documentation

- ☐ Maintain current contact details for recovery personnel.
- ☐ Define communication channels to use during an outage.
- ☐ Identify who communicates with employees, customers, vendors, and leadership.
- ☐ Keep DR procedures accessible even if normal systems are unavailable.
- ☐ Document roles and responsibilities.
- ☐ Document recovery priorities and decision-making authority.
- ☐ Record actions and decisions during an incident.
- ☐ Communicate during all outages, regardless of size.

15. Ongoing Review

- ☐ Review the DR plan after significant infrastructure or application changes.
- ☐ Review RPO and RTO targets periodically with business owners.
- ☐ Update dependency maps when systems change.
- ☐ Review backup and replication success regularly.
- ☐ Verify contact information remains current.
- ☐ Incorporate lessons from incidents and DR tests.
- ☐ Confirm the DR strategy continues to meet business requirements.

- ☐ Ensure disaster recovery planning remains aligned with business continuity planning.